HydroBoost – User Guide

1. Set the model input parameters in the Excel spreadsheets located in *Input_Spreadsheets/Model_A* (or Model_B / Model_C).

2. Save the updated spreadsheet in both locations:
   Input_Spreadsheets/Model_A (or Model_B / Model_C)
   Core_Models/HydroBoost/generated_data/Model_A (or Model_B / Model_C)

3. Run the Python scripts in the *Price_Forecasting* folder to refresh settings and network data. In each of *perfect_foreseight_model.py*, *mean_persistence_model.py*, and *import_data.py*, select the desired model (Model_A, Model_B, or Model_C) at the bottom of the script.

4. In the Julia script *HydroBoost_Sim.jl*, select the same model (Model_A, Model_B, or Model_C).

5. Start a Julia REPL (terminal).

6. In the Julia REPL, run the following (example with Model_C):

   ```
   import Pkg; Pkg.activate("."); Pkg.instantiate()
   include("Core_Models/HydroBoost/src/HydroBoost_Sim.jl")
   using .HydroBoost_Sim
   settings     = HydroBoost_Sim.read_ALEAF_HydroBoost_setting("Model_C")
   network_data = HydroBoost_Sim.generate_networkdata_HydroBoost(settings, "Model_C")
   result      = HydroBoost_Sim.execute_HydroBoost_model("Model_C")
   ```

7. Review the simulation outputs in the *Simulation_Results* folder.

8. Tip: use the same model name consistently across steps 3–6.

9. For any questions, doubts, or requests, please email me at agrimaldi@anl.gov